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Cognitive capacities, mental modules, and neural regions

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Dan Lloyd (2011) issues a salutary warning against the assumption of what I shall call *neural modularity* – the view that there is a one-to-one mapping between cognitive functions and distinct brain regions. He shows how the assumption can distort the interpretation of neuroimaging studies and blind researchers to global structures and activity patterns that may be crucial to many aspects of cognitive function and dysfunction.

In this note I want to add a further dimension to the discussion by making connections with the notion of *mental modularity* developed by evolutionary psychologists. What is the relation between mental and neural modularity? Do the arguments for massive mental modularity also support neural modularity? I shall offer some preliminary remarks on these questions and their bearing on issues in psychopathology.

Modern discussion of mental modularity takes its start from Fodor's *The Modularity of Mind* (1983). Fodor's conception of a mental module is a strict one, appropriate for peripheral systems, such as vision and language. Modules in his sense are special-purpose computational mechanisms which are to a large extent innate and hardwired, deliver shallow (non-conceptual) outputs, and operate in a way that is typically mandatory, fast, inaccessible, and informationally encapsulated. More recent writers adopt a more relaxed notion of mental modularity, which is applicable to central cognitive processes (e.g. Barrett and Kurzban 2006; Carruthers 2006; Coltheart 1999). Modules in this sense are functionally specialized mechanisms, which are dedicated to particular cognitive tasks, such as theory of mind, biological classification, and cheater-detection, and which can deliver conceptualized beliefs and desires as outputs. Their key property is domain-specificity, and it is an empirical question to what extent a given module possesses the other features on Fodor's list. It is arguable that evolution would have favoured the development of such central modules as well as peripheral Fodorian ones. Complex evolved systems are typically composed of specialized subsystems, and it is likely that specialized domain-specific cognitive mechanisms would be more efficient than general-purpose ones, especially given the wide range of adaptive problems faced by humans and other animals. Building on such arguments, some

theorists maintain that the mind is composed wholly, or mostly, of modular subsystems – a view known as the 'massive modularity hypothesis' (e.g. Carruthers 2006; Cosmides and Tooby 1992; Pinker 1997; Sperber 1994).

My aim here is not to defend massive modularity, but simply to ask whether mental modularity involves a commitment to neural modularity as well, and thus whether the arguments for the former carry over to the latter. It will be helpful to think in terms of three descriptive enterprises – personal psychology, subpersonal psychology, and neurology (compare Dennett 1981). Personal psychology is an idealized competence theory; it ascribes psychological traits to organisms on the basis of their behaviour. Talk of cognitive capacities is pitched at this level. Subpersonal psychology posits internal information-processing mechanisms that support personal-level psychological traits. The massive modularity hypothesis belongs to this level. Neurology, of course, deals with the fine-grained biological structures and processes in which the other levels are realized. Now, neural modularity is a claim about the relation between personal psychology and neurology; it is the claim that each cognitive capacity is realized in a localized brain region. And the question of whether massive mental modularity entails neural modularity breaks down into two sub-questions – one concerning the relation between personal and subpersonal psychology, and the other the relation between subpersonal psychology and neurology. The first question is whether each cognitive capacity is subserved by a single proprietary module; the second whether each module is realized in a distinct brain region. If the answers to both questions are positive, then mental modularity is equivalent to neural modularity. I take it that, for a given organism, the answers *might* be positive, but I shall argue that there is no reason to think that they *must* be, or even that it is particularly likely that they are – and thus that mental modularity does not *entail* neural modularity.

Consider first the relation between mental modules and neural regions. Some theorists suppose that mental modules will have localized neural bases and that evidence of neuroanatomical localization is required to support claims of mental modularity (Fodor 1983, pp.98-99; Panksepp and Panksepp 2001). But this is too strong. Mental modules are functionally characterized, and although there must be some neural structures that realize each module's processing mechanisms and knowledge base and establish its informational connections with other subsystems, these structures could be distributed across different neural regions (Segal 1996). (They could also change over time – for example, in response to damage.)

Conversely, the same neural region might be involved in supporting several different modules. Modules, like other complex biological systems, will probably be assembled from pre-existing components, and there may be considerable sharing of parts between modules (Carruthers 2006, ch.1). Thus, there need not be a one-to-one mapping between mental modules and neural regions, and indeed it seems unlikely that there will be.

What about the relation between cognitive capacities and mental modules? Should we expect each capacity to be subserved by its own dedicated module? If we think of cognitive capacities as abilities to solve particular adaptive problems, then the answer, I think, should be *no*. It may be true that the ideal architecture would be one with a dedicated subsystem for each task, but it is possible that modules could contribute to solving more than one problem (Samuels 2000), and given that evolution is frugal, preferring to adapt pre-existing mechanisms than to build new ones from scratch, we might expect to find a degree of multi-functionality within the human mind. Modules might also acquire new functions in the course of development, becoming modified to deal with novel tasks. For example, in literate societies children's object-recognition systems might develop a specialization for text processing (Barrett and Kurzban 2006).

As well as some modules having multiple functions, some functions may be subserved by a multiplicity of modules. A plausible example is that of conscious, flexible, domain-general reasoning, of the sort that involves working memory. (Currently popular 'dual-system' theories of mental architecture assign this to a distinct system, System 2; see Evans and Frankish 2009.) The existence of this sort of thinking presents a challenge for the massive modularity hypothesis, and in response some theorists have suggested that it is performed by a *virtual* system, supervening on the joint activity of many modules. A sophisticated version of this view has been developed by Peter Carruthers (2006, 2009). According to Carruthers, conscious thinking involves the mental rehearsal of actions, exploiting systems that originally evolved for the fine-grained control of action. The idea is that motor schemata are activated offline, generating perceptual feedback, which is then globally broadcast and taken as input by the various modular systems, thereby stimulating a new cycle of cognitive activity. Of particular importance, Carruthers argues, is the mental rehearsal of utterances, which may combine outputs from different modules. These generate feedback in the form of inner speech, whose content is extracted by the speech comprehension system and then globally broadcast and evaluated. Carruthers argues that these processes of rehearsal and

global broadcast would vastly extend a creature's problem-solving abilities, allowing it to evaluate potential actions, form novel thoughts, and develop new reasoning strategies.

This view has many attractions, and there are independent reasons for thinking of System 2 as a virtual one (Frankish 2009). But whether or not it is correct, it illustrates how sophisticated cognitive abilities could in principle emerge from the coordination of pre-existing modular subsystems – each itself perhaps distributed across multiple neural regions. Of course, this sort of distributed processing is different from that discussed by Lloyd, which involves connectionist networks. On a massively modular view there is functional discreteness at the subpersonal level, which is absent from radical connectionist models. (Moreover, the *internal* operations of modules are usually assumed to be computational rather than connectionist in character.)

There is no reason, then, to assume a one-to-one mapping between cognitive capacities and mental modules. It is worth noting that the points just made about the relation between cognitive capacities and mental modules would apply equally to the relation between cognitive capacities and *neural* modules, if there were such. Even if the brain had a modular structure, it would not follow that there was a one-to-one mapping between these modules and cognitive capacities identified at the personal level: some neural modules might support more than one capacity and some capacities might supervene on more than one module. (I have been using the term 'neural modularity' for the view that there *is* a one-to-one mapping between capacities and neural regions, but we can imagine a different kind of neural modularity, in which the functional taxonomy of neural regions is not derived from personal psychology.)

The moral is that massive modularists should heed Lloyd's warning. Even if the mind is massively modular, we should not make an assumption of neural modularity and should still use great caution in designing and interpreting imaging experiments. The implications for psychopathology should be obvious. Evolutionary psychology holds out the prospect of a new taxonomy of mental disorders in terms of the malfunctioning of specific cognitive subsystems (e.g. Murphy and Stich 2000). But for the reasons discussed, the relations between modular malfunction and brain activity on the one hand and cognitive capacities on the other are likely to be complex. Abnormalities in a single brain region might affect multiple modules, and abnormalities in different regions might affect just one module. Likewise, malfunctioning in one module might affect more than one cognitive

capacity, and malfunctions in different modules might affect the same cognitive capacity. Moreover, if something like Carruthers' account of conscious thought is correct, then impairments of working memory (which seem to be a component of schizophrenia) could be caused by many different types of modular malfunction. On Carruthers' view, working memory – the system responsible for the short-term storage and manipulation of information in conscious reasoning – is not a mental module, still less a neural one. Rather, it is a virtual system, which supervenes on the various feedback loops employed in mental rehearsal and on the suite of conceptual and sensory modules involved in generating action schemata and processing the feedback they produce (Carruthers 2006, p.141). Disruption to any of these subsystems could, in principle, cause disorders of working memory.

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